

3.8 Pilot Study

3.8.1 Purpose of Pilot Study

Prior to the main data collection, a pilot study was conducted with one team of three participants. The team was multicultural, consisting of a Dutch participant, a Chinese participant born and raised in the Netherlands, and a participant of Dutch-Chinese background born and raised in the United States. While all three participants had ties to Dutch and/or Chinese cultural backgrounds, their diverse lived experiences made them a suitable pilot group for testing whether the study design could surface intercultural dynamics. The pilot served three main purposes. First, it aimed to assess the procedural and logistical feasibility of the study design, including the scheduling, the flow of the session, the clarity of instructions given to participants, and the practical manageability of running both experimental conditions within a single session. Second, it tested whether the vignette tasks were effective in eliciting genuine conflict between team members, that is, whether the assigned roles and competing priorities were sufficiently distinct to produce disagreement and negotiation within the 15-minute timeframe. Third, the pilot provided an opportunity to evaluate the LLM-based cultural mediator’s performance in a live experimental setting, including the quality and relevance of its generated advice, the usability of the interface under real conditions, and whether the CSV output was produced as intended.

3.8.2 The Original Task Design

The original study design adopted a mixed-methods approach, combining quantitative and qualitative methods to examine team conflict in intercultural teams and the potential impact of an LLM-based intervention. Originally, between 60 and 90 participants were to be recruited through convenience sampling to form teams of three, organised into four blocks for logistical purposes. The design included two experimental conditions: a minimal control condition, in which teams resolved conflict without any intervention, and an LLM-mediated condition, in which teams received support from the cultural mediator. Using a within-subjects repeated-measures design, each team completed both conditions across two counterbalanced vignette tasks to control for order, learning, and fatigue effects.

The two vignette tasks were a community app feature selection task and a community festival planning task (see Appendix F). In the app task, teams had to select and rank four out of six available features, with each participant assigned a distinct user role (Practical User, Social & Motivated User, and Careful & Reflective User) to create conflicting priorities. In the festival task, teams had to select and rank four

out of six planning steps, with roles representing different stakeholder perspectives (Resident Representative, Event Organiser, and Youth Volunteer Leader). In both tasks, participants had 15 minutes to reach a shared decision, and were required to consult the LLM-based mediator during the discussion in the intervention condition.

The compensation structure followed a pay-for-performance principle, in which participants had a higher chance of winning a €10 voucher if they succeeded in advancing their strongly preferred option during the task, and a lower chance if the team failed to complete it. Post-tests included measures of personality (Big Five and Short Dark Triad), general attitudes towards AI, conflict type, team satisfaction, and responsibility attribution (the Short Dark Triad and responsibility attribution were later removed from the final design). The session concluded with a focus group.

3.8.3 Findings from the Pilot

The pilot study revealed findings across three areas: task design, mediator usage, and LLM mediator performance.

Task Design. Task I, the community app feature selection task, proved problematic in several respects. Participants found the goal of the app unclear, which undermined their ability to engage meaningfully with their assigned roles. As one participant noted during the session:

I think it's because now it's very vague. So I think it's difficult for me to think about reasons why I want my features when I don't understand why I want that feature.

The six available options were also too similar to one another to generate substantive disagreement, and roles lacked sufficient concrete reasoning material to anchor participants' positions. Overall, Task I did not reliably elicit the level of conflict necessary for the study's purposes. Task II, the community festival planning task, was received more positively. Participants found the goal concrete and the planning steps sufficiently interdependent to naturally produce disagreement over sequencing priorities.

Mediator Usage. Two procedural issues emerged regarding the LLM mediator. First, participants were unfamiliar with the concept of a mediator and uncertain about its function in the context of the task. Second, even once the tool was introduced, participants were unsure what to ask it, as illustrated by one participant's comment during the session: *"I'm confused on what questions we're supposed to ask the LLM."* The open-ended consultation format placed too much cognitive burden

on participants mid-task, and the absence of a fixed time point for consultation introduced variability in how and when the mediator was used. Additionally, participant roles had been communicated during the session via Teams chat rather than in advance, which disrupted the flow of the procedure.

LLM Mediator Performance. Focus group feedback indicated that participants found the mediator’s output relevant to the actual dynamics of their conversation rather than generic. However, one participant observed that while the mediator supported the quality of the discussion, it did not directly assist with the objective of ranking the planning steps, which highlights an important boundary between facilitating communication and supporting task completion. Overall, participants considered the mediator helpful within its intended role.

3.8.4 Task Refinement

Based on the findings of the pilot, several refinements were made to the task design. The most substantial change concerned Task I, the community app feature selection task. Pilot observations indicated that the goal of the app was unclear to participants, that the six available options were too similar to one another to generate meaningful conflict, and that the roles lacked sufficient concrete reasoning material to anchor participants’ positions. As a result, Task I was removed from the design entirely. The number of options was reduced from six to three across the remaining task, and roles were reoriented around practical stakeholder priorities rather than user emotions, with example reasoning added to support participants in articulating their positions.

Task II, the community festival planning task, was retained but refined. Pilot participants found this task considerably clearer, as the goal was concrete and the planning steps were interdependent, which naturally elicited conflict over sequencing priorities. The task was narrowed from a broad community festival to a one-day music festival to increase specificity and reduce ambiguity. The number of planning steps was reduced from six to three, and roles were restructured to represent distinct departmental perspectives within a team.

Several procedural adjustments were also made to the mediator usage. In the original design, participants could consult the LLM mediator as many times as they wished; this was changed to a single mandatory consultation at a fixed time point, to standardise exposure across teams and reduce uncertainty about when and how to use the tool. Participants in the pilot reported confusion about what to ask the mediator, so default prompt options were introduced as guidance. Additionally, participant roles had been communicated via Teams chat during the session; this

was changed so that roles and planning steps were sent to participants in advance of the session to allow adequate preparation.

3.8.5 Pre-Experimental Validation

The cultural mediator’s answers were tested on consistency in several rounds. During these tests, the prompts were changed and improved. In addition, five example transcripts with correct labels were created to check whether the CSV output on all relevant constructs (pragmatic failure, value misalignment and conflict types) was accurate. The cultural mediator correctly classified four out of five cases. The only mistake happened in a case where the differences between categories were very subtle and therefore also difficult for humans to distinguish.

3.9 Ethical Considerations

An Ethics and Privacy Quick Scan was completed in accordance with Utrecht University’s terms and conditions, which classified the project as low risk. Prior to participation, all participants provided informed consent via a Qualtrics consent form, which included an explanation of their right to withdraw at any point without consequence. Audio recordings were used solely for transcription purposes.

F Pilot Task Vignettes

F.1 Task I: Deciding the First Version of a Community App

The clock is ticking.

You’ve just joined a small product team preparing to launch a new mobile app. Six features are on the table, but management has made the constraint clear: the first release can include only four. Your team must decide which features make the cut—and which are left behind.

“We need something that works offline.”

“People won’t use it if it isn’t fun.”

“Shouldn’t we focus on useful insights instead of gimmicks?”

Each team member speaks from a different user perspective, with their own needs and priorities. Trade-offs are unavoidable, and disagreements about what matters most are expected. There is no single correct solution—the goal is to discuss, negotiate, and reach a shared decision.

You have 15 minutes.

Teams must:

1. Select exactly four features, and
2. Rank them in priority order from one (highest) to four (lowest)

If agreement is not reached within the time limit, the task is considered incomplete.

AI-based mediator (Condition II). During the discussion, teams in the intervention condition must consult an AI-based mediator one time. The mediator can summarise statements, highlight disagreements or trade-offs, invite quieter members to contribute, and help structure the discussion. The mediator does not recommend features or make decisions; final responsibility remains with the team.

Available App Features

- **Real-time chat** – users can message each other instantly
- **Rewards and badges** – points, streaks, and achievements

- **Activity insights** – statistics about usage and habits
- **Personal profiles** – customisable pages and preferences
- **Offline mode** – app functions without internet access
- **Smart suggestions** – personalised automatic recommendations

Assigned User Roles

Each participant adopts one role representing a specific user perspective.

Table 21. Participant roles and preferences

Role	Main Priority	Strongly Prefer	Refuse	Rationale
Practical User	Reliability & convenience	Offline Mode, Real-time Chat	Rewards & Badges	Wants the app to work anywhere and support easy communication; gamification feels distracting.
Social & Motivated User	Fun & engagement	Rewards & Badges, Personal Profiles	Smart suggestions	Motivation and personalisation increase enjoyment; statistics feel boring.
Careful & Reflective User	Understanding & guidance Clarity & thoughtful support	Activity Insights, Smart Suggestions	Real-time Chat	Prefers useful information and guidance; constant messaging feels overwhelming.

Task. Your team must select four features and agree on their priority order. Each member should explain their reasoning and participate actively. Disagreement is expected.

F.2 Task II: Planning a Community Festival

The clock is ticking.

You’ve just joined a small decision-making team tasked with planning the town’s community festival. Six key planning steps are on the table, but due to time and resource constraints, only four can be prioritised for immediate focus. Your team must decide which steps matter most—and in what order—to ensure the event is successful, safe, and engaging.

“Without a venue, nothing else matters.”

“But if we don’t promote it early, no one will show up.”

“Shouldn’t we plan the fun parts first?”

Each team member represents a different stakeholder perspective, bringing their own priorities and concerns. Trade-offs are unavoidable, and disagreements about what matters most are expected. There is no single correct answer—the goal is to discuss, negotiate, and reach a shared decision.

You have 15 minutes.

Teams must:

1. Select exactly four planning steps, and
2. Rank them in priority order from one (highest) to four (lowest)

If agreement is not reached within the time limit, the task is considered incomplete.

AI-based mediator (Condition II). During the discussion, teams in the intervention condition must consult an AI-based mediator one time. The mediator can summarise statements, highlight disagreements or trade-offs, invite quieter members to contribute, and help structure the discussion. The mediator does not recommend solutions or make decisions; final responsibility remains with the team.

Available Planning Steps

- Secure the festival venue
- Set the festival date and schedule
- Promote the festival to the community
- Arrange entertainment / activities
- Organise food and vendors

- Plan safety and logistics (permits, insurance, security, first aid)

Assigned User Roles

Each participant adopts a role representing a stakeholder perspective.

Table 22. Stakeholder roles

Role	Main Priority	Strongly Prefer	Refuse	Rationale
Resident Representative	Inclusion & attendance	Promote the festival, Set date & schedule	Plan safety & logistics first	Wants early awareness and clear timing. Administrative planning feels premature without community engagement.
Event Organiser	Logistics & feasibility	Secure venue, Plan safety & logistics	Arrange entertainment & activities first	Without a venue and safety, the festival can't happen. Fun elements feel secondary.
Youth Volunteer Leader	Fun & engagement	Arrange entertainment & activities, Organise food & vendors	Plan safety & logistics first	Exciting activities and food attract participants and volunteers. Administrative planning slows momentum.

Task. Your team must select four planning steps and agree on their priority order. Each member should explain their reasoning and participate actively. Disagreement is expected.